TEK ZANITH

Senior QA Automation Engineer | AI & Test Architecture Lead

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AI-native Senior Quality Assurance Engineer and Test Architect with 4+ years of experience leading end-to-end quality engineering across distributed web, mobile native, and enterprise systems. Specialized in **multi-agent test orchestration**, **Playwright & Maestro test automation**, and **code-level AST defect isolation** across high-concurrency fintech, e-government, and B2B SaaS platforms. Proven track record eliminating release bottlenecks by replacing manual testing with autonomous swarms, multi-device mobile validation fleets, and zero-leak Row Level Security (RLS) enforcement.

CORE COMPETENCIES & TECHNICAL STACK

TEST AUTOMATION & ENGINEERING

Playwright Automation, Maestro CLI, Selenium WebDriver (Python/C#), Bun Test Runner, Postman API Testing, Regression Suites, E2E Workflow Architecture.

AI & AGENTIC ORCHESTRATION

Multi-Agent Testing Swarms, Model Routing (Opus, Sonnet, Codex), Prompt Engineering, Deterministic Mocking, Synthetic Data, AST Bug Isolation.

MOBILE & MULTI-PLATFORM QA

React Native & Expo, Android (API 36/37) & iOS Fleets, Foldable Viewports, Compact/Flagship Device Extremes.

PERFORMANCE, SECURITY & CLOUD

Supabase Row Level Security (RLS), PostgreSQL, Apache JMeter, Locust Load Testing, WebSockets, Webhooks, SSO Authentication.

DEVOPS, CI/CD & REPORTING

GitLab CI/CD, GitHub Actions, Jenkins, Standalone Zero-Dependency HTML Reports (Inline Base64), Automated Two-Way Test Tracking (Google Sheets API, Trello), Shift-Left Quality Gates.

PROFESSIONAL EXPERIENCE

Senior QA Automation Engineer & AI Test Architect · Skai Technology

06/2026 – PRESENT

(Phnom Penh, Cambodia)

Directing test automation architecture and quality engineering across high-velocity B2B SaaS, mobile native, and enterprise financial platforms.

- **Autonomous Multi-Agent Test Swarms:** Architected a 3-agent specialist swarm (API/Security, Realtime/Webhooks, UI/Workflow) running parallel validation across multi-tenant B2B SaaS platforms, cutting full regression verification cycles by 65%.
- **Zero-Leak Authorization & Security:** Eliminated cross-tenant authorization vulnerabilities by designing validation suites for Supabase Row Level Security (RLS) and additive role pyramids, guaranteeing 100% financial and billing data isolation.
- **Deterministic Web & API Automation:** Automated 52 critical live-data user journeys in Playwright with a 100% pass rate, embedding 98 automated visual assertions and verifying real-time WebSocket sync latencies under 3 seconds.
- **Cross-Platform Mobile Fleet Architecture:** Engineered cross-platform mobile test architecture for React Native/Expo apps across 5 extreme device profiles (foldable viewports, compact Android, and flagship iOS/Android), resolving responsive coordinate clipping before production releases.
- **Mobile Pipeline Speed & AST Bug Isolation:** Implemented Maestro CLI automation reducing smoke test runs from 4 hours to 18 minutes, and diagnosed defects down to the AST and component source code level to accelerate developer fix velocity.
- **Enterprise Load Testing & Standalone Reporting:** Executed Locust and JMeter load benchmarks under high-concurrency SSO traffic, and engineered zero-dependency HTML audit reports with inline Base64 proofs for client and executive sign-off.

PROFESSIONAL EXPERIENCE (CONTINUED)

Quality Assurance Engineer (Sole Lead) · Digital Economy and Business Committee

05/2024 - 01/2026

(Phnom Penh)

Spearheaded quality assurance and release engineering for national digital economy platforms and e-government portals.

- **Institutional QA Infrastructure:** Established the organization's foundational QA infrastructure, authoring Standard Operating Procedures (SOPs), bug classification matrices, and release sign-off gates adopted across all engineering teams.
- **Concurrency & Stress Engineering:** Identified concurrency bottlenecks prior to public rollouts by engineering high-volume stress and load testing suites in Apache JMeter, ensuring server stability under heavy public traffic spikes.
- **AI-Assisted Efficiency:** Accelerated test suite authoring by 50% by designing prompt-engineering pipelines with Claude and ChatGPT to synthesize complex test matrices, edge-case permutations, and API payloads.
- **Zero Critical Escapes:** Maintained a 0% critical bug escape rate across production releases through rigorous manual, API, and cross-browser regression gates.
- **Executive Alignment:** Aligned requirements between developers, product owners, and government executives through structured, data-backed daily release health dashboards.

Quality Assurance Engineer · Infinity Information and App Dev (Phnom Penh)

01/2024 - 04/2024

Delivered manual and API quality validation for client web and mobile applications under rapid two-week sprint cycles.

- **API Contract Verification:** Prevented contract-breaking API changes by building comprehensive Postman test collections, validating JSON schemas, status codes, and latency across all backend services.
- **Comprehensive Test Documentation:** Authored 200+ detailed, reusable test cases covering positive, negative, boundary, and edge-case scenarios accessible to both technical and client stakeholders.
- **Triage Optimization:** Shortened defect triage lifecycles by 30% by submitting reproducible defect reports containing network logs, payload dumps, and environmental parameters.

Quality Assurance Engineer · Techbodia (Phnom Penh)

03/2022 - 09/2022

Developed test automation frameworks and executed performance engineering for enterprise software systems.

- **Automated Regression Testing:** Automated repetitive regression testing by implementing Selenium WebDriver (Python) and Playwright automation frameworks, cutting manual testing effort by 40%.
- **Response Boundary Benchmarking:** Benchmarked system response boundaries using Apache JMeter, identifying database connection pool exhaustion and memory leaks under simulated load conditions.
- **Agile Collaboration:** Integrated automated testing suites into team sprint cadences, participating in Agile ceremonies and executing rapid post-fix verification.

PROFESSIONAL EXPERIENCE (CONTINUED)

Quality Assurance Engineer / Project Lead · Kirirom Institute of Technology

02/2022 - 10/2022

(Kampong Speu)

Led QA operations for computer vision systems and client SDK initiatives, reporting directly to executive leadership.

- **Computer Vision Platforms:** Led QA verification for mission-critical computer vision platforms, including Face Recognition Attendance Systems, License Plate Recognition (LPR) platforms, and client-facing Mobile SDKs.
- **Automated CI/CD Gates:** Built automated CI/CD quality gates using GitLab CI/CD, Jenkins, and Google Cloud Platform (GCP), preventing broken builds from progressing to staging.
- **Executive Milestone Reporting:** Reported weekly milestone velocity directly to the CTO, translating technical defect metrics into clear risk assessments and release decisions.

Learning Management System (LMS) Operator · Kirirom Institute of Technology

02/2022 - 05/2022

- Administered platform workflows, user access controls, and lesson delivery pipelines for faculty and students, ensuring 99.9% LMS availability during exam periods.

ARCHITECTURAL HIGHLIGHTS & METHODOLOGICAL CASE STUDIES

Multi-Agent Testing Swarm & Model-Routing Architecture

Designed an autonomous testing hierarchy pairing task-specific LLM models: Claude Opus for UI component state validation, Claude Sonnet for mechanical screen translation, and OpenAI Codex for backend business rules, DB migrations, and Maestro concurrency tests. Backed by cross-model automated peer reviews and deterministic synthetic test data generation.

Cross-Platform Extreme Form-Factor Mobile Validation Fleet

Established an automated mobile testing fleet evaluating five distinct form-factor extremes (Pixel Fold, Pixel 10 Pro XL, iPhone SE, and iPhone 17 Pro Max). Leveraged Maestro CLI YAML suites to automate user journeys, isolating layout coordinate clipping and view hierarchy collapses directly in React Native and Expo component code.

EDUCATION & LEADERSHIP

Bachelor of Science in Software Engineering
Kirirom Institute of Technology (KIT)
Kampong Speu, Cambodia

Leadership Program Short Course
University of the Thai Chamber of Commerce (UTCC)
Bangkok, Thailand

ACHIEVEMENTS & LANGUAGES

- 1st Runner-Up — AWS Build on Cambodia Hackathon (2021)
- Participant — SmartStart & SmartSpark Innovation (2021)
- Finalist — Techno Innovation Cambodia Hackathon (2021)
- IELTS 6.5 — Academic English Certification (2018)

Khmer: Native | **English:** Advanced (IELTS 6.5) | **French:** Intermediate | **Japanese:** Elementary